

ALEX MONITO NHANCOLOLO

PhD Student of Statistics and Probability

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Last modified: April 24, 2026

PERSONAL PROFILE

A. M. Nhancololo is a Ph.D. student in Statistics and Probability at IME-USP and a visiting researcher at Jaume I University (2026). He holds an M.Sc. from UFLA (Best Thesis Award, 2024) and a B.Ed. from Universidade Save. Currently, he is pursuing an M.Sc. in Financial Engineering at WorldQuant University and an MBA in AI and Big Data at ICMC-USP. Awarded third place in the 2024 Spatial AI Challenge (I-GUIDE), his research involves spatio-temporal statistics applied to social, environmental, and medical sciences. Methodologically, he focuses on geospatial intelligence (GeoAI), spatial point processes, lattice/area data, and geostatistics.

EDUCATION

Ph.D. 2028 [*]	Institute of Mathematics and Statistics, University of São Paulo, Brazil Ph.D. in Statistics and Probability. Academic Transcript Thesis: To be defined Advisor: Airlane Pereira Alencar (Statistics)
M.S. 2026 [†]	WorldQuant University, USA M.Sc. in Financial Engineering. Academic Transcript
M.S. 2023	Institute of Exact Sciences and Technology, Federal University of Lavras, Brazil M.Sc. in Statistics and Experimental Agriculture. Academic Transcript Thesis: Univariate spatial point processes applied to the distribution of tree species in natural forests. Institutional Repository Advisor: João Domingos Scalon (Statistics & Spatial Statistics)
B.S. 2021	Save University, Mozambique B.Ed. in Mathematics and Statistics. Academic Transcript Thesis: Analysis of the Potential Use of Three-Dimensional Models in Teaching Geometric Solids Advisor: André Silvestre Cuinica (Mathematics, Meteorology & Physics)

SUPPLEMENTARY EDUCATION

2026 ²	MBA in Artificial Intelligence and Big Data at the Institute of Mathematics and Computer Sciences (ICMC) – University of São Paulo (USP)
2024	Data Analysis with Python (Duration: 73 hours), University of São Paulo (ESALQ)
2024	Applied AI Lab: Deep Learning for Computer Vision (Duration: 16 weeks), WorldQuant University, USA
2024	Applied Data Science Lab (Duration: 16 weeks), WorldQuant University, USA
2023	Applied Geostatistics to Precision Agriculture (Duration: 20 hours), State University of Campinas, UNICAMP, Brazil
2023	Introduction to Precision Agriculture and its Technologies (Duration: 16 hours), State University of Campinas, UNICAMP, Brazil
2017	Poultry Farming (Duration: 32 hours)

^{*}Expected

[†]Expected

²Expected

2017 Electrical Installation (Duration: 32 hours)

PROFESSIONAL APPOINTMENTS

12/01–12/02/2026 Teacher, Summer Course: Introduction to Spatial Statistics (30 hours), [IME-USP Website](#), [Book/Course Material](#) and [Personal Website](#)

01/07/2025 (current) Teaching Assistant of Introduction to Data Analysis (MAE0261), University of São Paulo (Supervisor: Prof. Airlane Pereira alencar)

01/01-2025 (current) Teaching Assistant for Introduction to Data Science (MAE5905), University of São Paulo (Supervisor: Prof. Pedro A. Morettin), [MAE5905 in Teaching section](#)

2022 Data Analyst, [GALP Total Energy](#)

2020-2021 High School Mathematics Teacher, Chongoene Secondary School

2018–2021 Teaching Assistant for Algebra I & II, [Save University](#) (Supervisor: Prof. André Silvestre Cuinica)

2018–2020 Teaching Assistant for Integral Calculus and Euclidean Geometry, [Save University](#) (Supervisors: Prof. José Portez Sibinde & Eduardo G. Tiago Macie)

AWARDS & HONORS

2025 **2024 Best Dissertation Award**, Graduate Program in Statistics and Experimental Agriculture, Federal University of Lavras. [Link](#)

2025 **Third Place**, Spatial AI Challenge 2024: AI-ready spatial data, machine learning models, and applications, I-GUIDE (Institute for Geospatial Understanding). [Link](#)
Title: Pipeline for Roof Material and Geometry Classification
Team: Julian Huang (University of Chicago) | Alex Monito Nhancololo (University of São Paulo) | Yue Lin (University of Chicago)

2025 **Certificate, Foundations of Financial Engineering**, WorldQuant University. [Link](#)

PUBLICATIONS

Publications

1. Victor, A.; Nhancololo, A. M.; Bah, H. A. F.; Xavier, S. P.; Gotine, A. R. M. E.; Moreira, V. A.; Charles, C. M.; Cerqueira-Silva, T.; Rondó, P. H. National and regional temporal trends and forecasting of preterm birth in Brazil: Evidence from national Birth data (2014–2023) with projections to 2030. *BMC Pregnancy and Childbirth*, 2025. DOI: [10.1186/s12884-025-08536-6](#)
2. Oliveira, W. A.; Scalon, J. D.; Nhancololo, A. M. Characterization of the spatial distribution of *Copaifera langsdorffii* and *Xylopia brasiliensis* in an urban fragment of the Atlantic Forest domain. *Ciência Rural*, 2025. DOI: [10.1590/0103-8478cr20240132](#)
3. Nhancololo, A. M.; Scalon, J. D.; Pereira, F. A. C.; Oliveira, W. A.; Silva, B. M. Comparison between the laboratory method and the Stolf penetrometer in soil density analysis: a study using geostatistical methods. *SIGMAE*, vol. 13, Jan. 2024. DOI: [10.29327/2520355.13.1-7](#)
4. Nhancololo, A. M.; Cuinica, A. S. Analysis of the Potential of Three-Dimensional Models in Teaching and Learning Spatial Geometry in 9th Grade: A Case Study of Chongoene Secondary School. *Kavenye Scientific Journal*, vol. 1, pp. 44–60, 2023. [\[Link\]](#)

Accepted Papers

5. Paludo, G. F.; Nhancololo, A. M.; Sousa Bueno Filho, J. *petioleFborder*: an R package with algorithms for automatic detection of petiole from the coordinates of leaf contour. *Plant Biosystems*, 2025. [\[Link\]](#)
6. Damaceno, G. P.; Paludo, G. F.; Nhancololo, A. M.; Scalon, J. D.; Sabe, E. M.; da Silva, G. F.; Bueno-Filho, J. S. S.; da Silva, D. M. Behavior of regression models for area data with outliers: an application to tuberculosis incidence. *Sigmae*, 2025. [\[PT paper Link\]](#)

BOOKS & LECTURE NOTES

1. Nhancololo, A. M.; Scalon, J. D.; Alencar, A. P. *Introdução à Estatística Espacial: Teoria e Aplicações com R*. Zenodo, 2026. DOI: [10.5281/zenodo.19256838](https://doi.org/10.5281/zenodo.19256838). [Interactive Course Material](#)

WORKING PAPERS

1. Nhancololo, A. M.; Mateu, J.; Paludo, G. F.; Alencar, A. P.; Rahman, A. A.; Scalon, J. D. Robust Bayesian Spatio-Temporal Modeling of Conflict Data. submitted, 2026.
2. Nhancololo, A. M.; Kawaichi, V. M.; Mateu, J.; Alencar, A. P. MGTWR.
3. Nhancololo, A. M.; Mateu, J.; Alencar, A. P. Log-Gaussian Cox Processes.
4. Nhancololo, A. M.; Alencar, A. P.; Cuinica, A. S.; Paludo, G. F.; Damaceno, G. P.; Lhone, M. A. Just-in-Time Repression: Modeling Police Violence Against Civilians in Mozambique (1997–2025) with Generalized Additive Models. submitted, 2025.
5. Nhancololo, A. M.; Paludo, G. F.; Gianasi, F. M.; de Oliveira, W. A. Spatial patterns of native tree species of an Atlantic Rainforest fragment and its association to edaphic variables. (**Based on the author's Master's thesis**) submitted.
6. Paludo, G. F.; Nhancololo, A. M.; Mendes, A. M.; Cayo, A. C.; Huacani, R. C.; Damaceno, G. P.; Filho, J. S. B. A memory game that explores variability. submitted, 2025.

TOOLS & SOFTWARE

- **Statistical Computing & Programming:** R, Python, SQL, SPSS, SISVAR
- **Productivity Tools:** Microsoft Office, LibreOffice

PRESENTATIONS

TALKS

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| 2025 | Nhancololo, A. M.; Alencar, A. P. “The Dynamics of Insurgent Hotspots in Cabo Delgado, Mozambique: A Bayesian Spatio-Temporal Analysis (2017–2024)”. Presented on November 11, 2025, at the 6 th Graduate Meeting in Statistics at IME-USP, São Paulo, Brazil. Link |
| 2025 | Nhancololo, A. M. “Terrorism in Cabo Delgado, Mozambique: A spatio-temporal analysis of fatalities and their determinants (2017–2024) with forecasts to 2030.” Presented on August 8, 2025, at the 69 th Brazilian Region of the International Biometric Society (RBras) and the 21 st Symposium on Statistics Applied to Agricultural Experimentation (SEAGRO), Vitória, Espírito Santo. Slides link |
| 2023 | Nhancololo, A. M.; Scalon, J. D. “Textual Data Analysis Using Natural Language Processing (NLP) in R.” (Seminar). Online Access: Link |

- 2024 Nhancololo, A. M.; *et al.* “Geostatistical Analysis of Soil Resistivity for Cassava Cultivation – Lago Azul Farm in the Pontal do Paranapanema Region, São Paulo (SP).” XXXIII Postgraduate Congress of UFLA (Conference). [Link](#)
- 2024 Nhancololo, A. M.; Scalon, J. D.; Oliveira, W. A.; Ossifo, M. E. “Characterization of the Spatial Pattern of Fire Incidents in the Municipality of Lavras, MG.” Brazilian Region of the International Biometric Society (RBras). [Link](#)
- 2024 Nhancololo, A. M.; Scalon, J. D.; Oliveira, W. A. “Spatial Modeling of Point Processes Applied to Forest Ecology: A Case Study with *Siparuna guianensis*.” Brazilian Region of the International Biometric Society (RBras). [Poster link](#)
- 2023 Nhancololo, A. M.; Scalon, J. D.; Oliveira, W. A. “Estimation of Bivariate Models for Point Processes with Spatial Attraction in Native Species.” XXXII Postgraduate Congress of UFLA. [Link](#)
- 2021 Nhancololo, A. M. “Use of the Fundamental Principle of Counting in the Teaching and Learning Process of Combinatorial Analysis.” UniSave Scientific Journeys. [Link](#)
- 2020 Nhancololo, A. M. “Exploration of the Mathematical Knowledge Involved in the Activity of a Gravedigger.” UniSave Scientific Journeys. [Link](#)
- 2019 Nhancololo, A. M. “Teaching Linear Equation Systems through Problem Solving.” UniSave Scientific Journeys. [Link](#)

POSTERS

- 2025 Nhancololo, A. M.; Macamo, A. M.; Cuinca, A. S.; Alencar, A. P.; Lhone, M. A. *Spatio-Temporal Variability of Precipitation in the Limpopo River Basin (1984–2023)*. [poster link](#)
- 2025 Nhancololo, A. M.; Cuinca, A. S.; Pereira, A. A.; Lhone, M. A. *Analysis of police violence against civilians in Mozambique (1994–2025)*. [poster link](#)
- 2025 Damaceno, G.; Paludo, G. F.; Nhancololo, A. M.; Sabe, E.; Silva, G. F.; Bueno Filho, J. S. S.; Silva, D. M.; Scalon, J. D. “Comparison of spatial statistical models for areal data with outliers.” *69th RBras and 21st SEAGRO*, Vitória, Espírito Santo. [Home page link](#), [Poster link](#)
- 2025 Oliveira, W. A.; Elias, R. S.; Barros, M. F.; Rosa, L. F.; Nhancololo, A. M.; Nascimento, E. S.; Nakamura, L. R.; Scalon, J. D. “Impact of social indicators on infant mortality: An approach using distributional models.” *69th RBras and 21st SEAGRO*, Vitória, Espírito Santo. [Home page link](#), [Poster link](#)
- 2025 Oliveira, W. A.; Barros, M. F.; Hoffmann, Y. T.; Rosa, L. F.; Nhancololo, A. M.; Nakamura, L. R.; Scalon, J. D. “Incorporating GMRF-based distributional models for analyzing violence against women in Minas Gerais.” *69th RBras and 21st SEAGRO*, Vitória, Espírito Santo. [Home page link](#), [Poster link](#)
- 2025 Paludo, G. F.; Mendes, A. M. F.; Damaceno, G.; Cayo, A. W. C.; Nhancololo, A. M.; Bueno Filho, J. S. S. “A memory game that explores variability.” *69th RBras and 21st SEAGRO*, Vitória, Espírito Santo. [Home page link](#), [Poster link](#)
- 2024 Scalon, J. D.; Oliveira, W. A.; Nhancololo, A. M.; Cordeiro, L. L.; Rosa, L. F. “Structured Space-Dependent Calibration Model.” Brazilian Region of the International Biometric Society (RBras).
- 2023 Nhancololo, A. M.; Scalon, J. D. “Comparison between the Laboratory Method and Stolf Penetrometer in Soil Density Analysis: A Study Using Geostatistical Methods.” Mineiro Statistics Meeting (MGEST).
- 2023 Oliveira, W. A.; Scalon, J. D.; Nhancololo, A. M. “Exploring Methodologies for Analysing Non-Stationary Point Processes: An Application in Forest Data.” XXXII Postgraduate Congress of UFLA.

PARTICIPATION IN EVENTS, CONGRESSES, EXHIBITIONS, AND FAIRS

- 2025 6th Graduate Meeting in Statistics at IME-USP, São Paulo, Brazil

2025	17th Minas Gerais Meeting of Statistics (MGEST), Minas Gerais, Lavras
2025	69 th Brazilian Region of the International Biometric Society (RBras) and the 21 st Symposium on Statistics Applied to Agricultural Experimentation (SEAGRO), Vitória, Espírito Santo.
2024	Rbras – Brazilian Region of the International Biometric Society (Congress) - Piracicaba, São Paulo 16th MGEST – Mineiro Statistics Meeting (Congress), Juiz de Fora, Minas Gerais XXXII UFLA Postgraduate Congress (Congress) - Lavras, Minas Gerais Teaching Methods for the Bachelor’s Degree in Statistics (Workshop) - Lavras, Minas Gerais IV Workshop in Data Science (Workshop) - Lavras, Minas Gerais V Northeast Forum of Statistics and Data Science (Workshop) XIV Summer Program of DES/ICET – UFLA (Workshop) - Lavras, Minas Gerais
2023	Conversation Club – B1 (Workshop); Writing Workshop – B1 (Workshop); Conversation Club – A2 (Workshop); Reading Workshop – A2 (Workshop) Corporate Education – The New Tool for Professional Advancement (Exhibition) II South Minas Forum of Municipal Councils for Racial Equality Policies & Related Entities – UFLA (Forum) Challenges and Strategies for Promoting Racial Equality in South Minas (Symposium) UniSave Scientific Journeys (Series of Workshops)
2018–2021	UniSave Scientific Journeys (series of student-led seminars)

EVENT ORGANIZATION

- 2023 – 6th UFLA Open Doors (Exhibition)
- Nhancololo, A. M.; Silva, D. M.; Pereira, G. M. C. *XVI Summer Program DES-ICET/UFLA*. 2025.

ACADEMIC REFEREES

Jorge Mateu Universitat Jaume I, Spain	Professor, Department of Mathematics mateu@uji.es
Airlane Pereira Alencar University of São Paulo	Professor, Department of Statistics lane@ime.usp.br
João Domingos Scalón Federal University of Lavras	Professor, Department of Statistics scalón@ufla.br
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